16 Nov

17:00

Reforzando la seguridad de Kubernetes con gVisor y Falco

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Sysdig

KCD SPAIN

/Me

1. Security
   a. +6 years as researcher
   b. Security content engineer at Sysdig
2. Talks / Events
   a. RootedCon, NavajaNegra, NoConName, Cybercamp, Codemotion CCN-CERT…
   b. Secrypt, HITB, KubeCon EU, DragonJar…
3. Code
   a. GraFScan
   b. DeepConfusables , fakevideoforensics (co-author with jiep)
   c. SpyCrap (co-author with ruthgnz)

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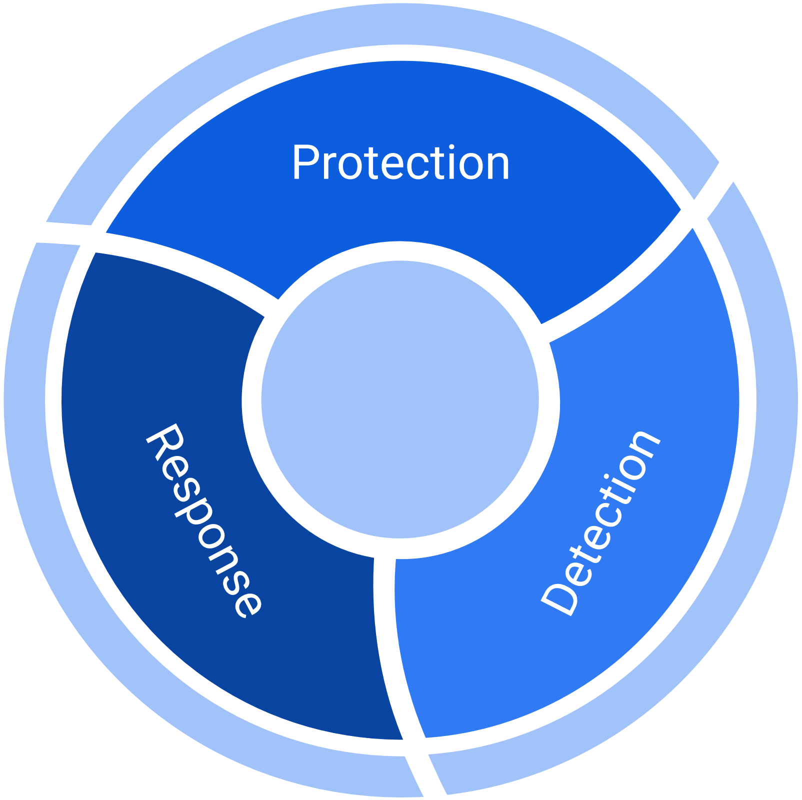
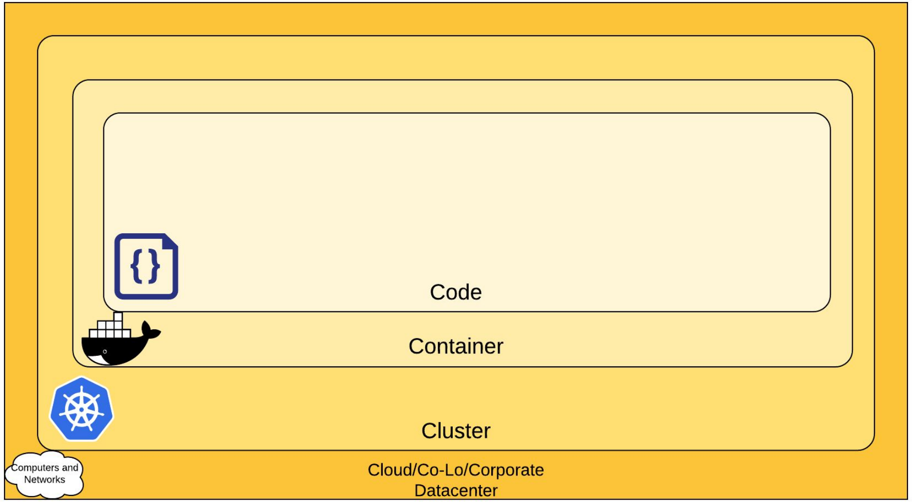
Agenda

- Kubernetes Security
- Sandbox containers
- Threat detection
- Conclusión

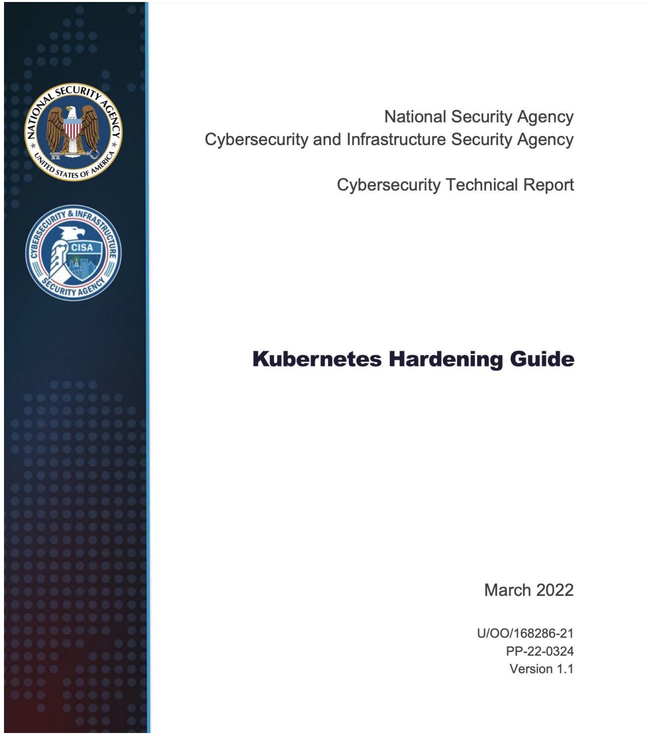
Security Strategy



Kubernetes Security Overview



https://kubernetes.io/docs/concepts/security/overview/



National Security Agency
Cybersecurity and Infrastructure Security Agency

Cybersecurity Technical Report

Kubernetes Hardening Guide

March 2022

U/OO/168286-21
PP-22-0324
Version 1.1

https://media.defense.gov/2022/Aug/29/2003066362/-1/-1/0/CTR_KUBERNETES_HARDENING_GUIDANCE_1.2_20220829.PDF

Kubernetes Security Overview

Cloud

Cloud provider security

If you are running a Kubernetes cluster on your own hardware or a different cloud provider, consult your documentation for security best practices. Here are links to some of the popular cloud providers' security documentation:

IaaS Provider	Link
Alibaba Cloud	https://www.alibabacloud.com/trust-center
Amazon Web Services	https://aws.amazon.com/security
Google Cloud Platform	https://cloud.google.com/security
Huawei Cloud	https://www.huaweicloud.com/securecenter/overallsafety
IBM Cloud	https://www.ibm.com/cloud/security
Microsoft Azure	https://docs.microsoft.com/en-us/azure/security/azure-security
Oracle Cloud Infrastructure	https://www.oracle.com/security
VMware vSphere	https://www.vmware.com/security/hardening-guides

Cluster

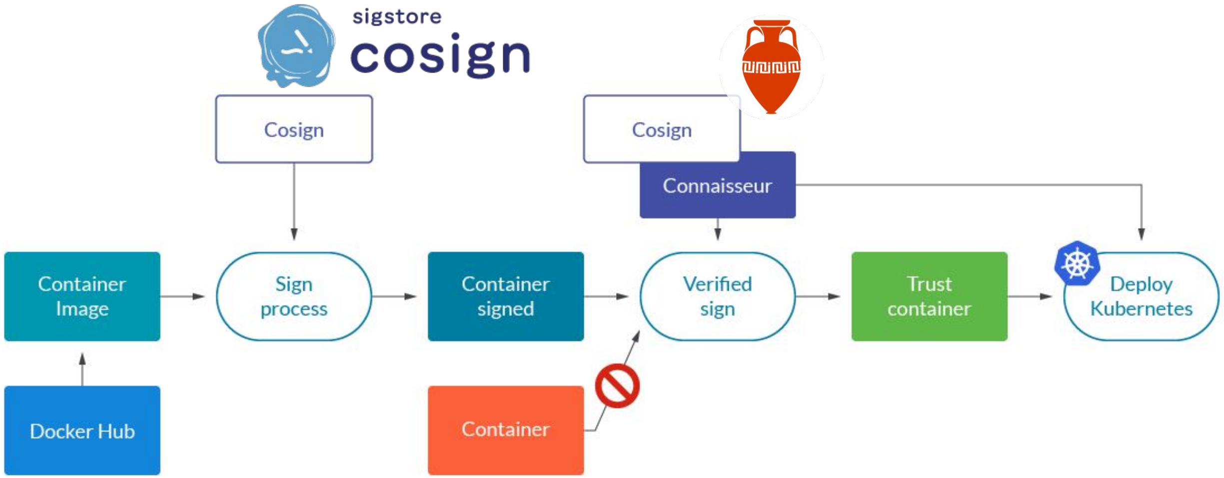
- Securing the cluster components that are configurable.
- Securing the applications which run in the cluster.
- RBAC Authorization
- Authentication
- PSP/OPA
- Network Policies

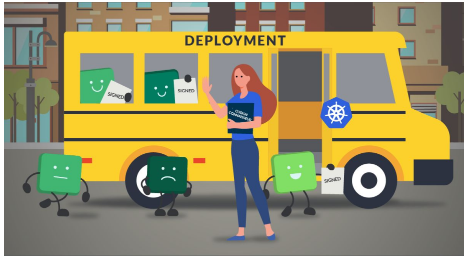
Container

- Container Vulnerability Scanning and OS Dependency Security
- No root privileges
- Signature
- Isolation

Kubernetes security - Signature

Trusted containers





https://sysdig.com/blog/secure-kubernetes-deployment-signature-verification/

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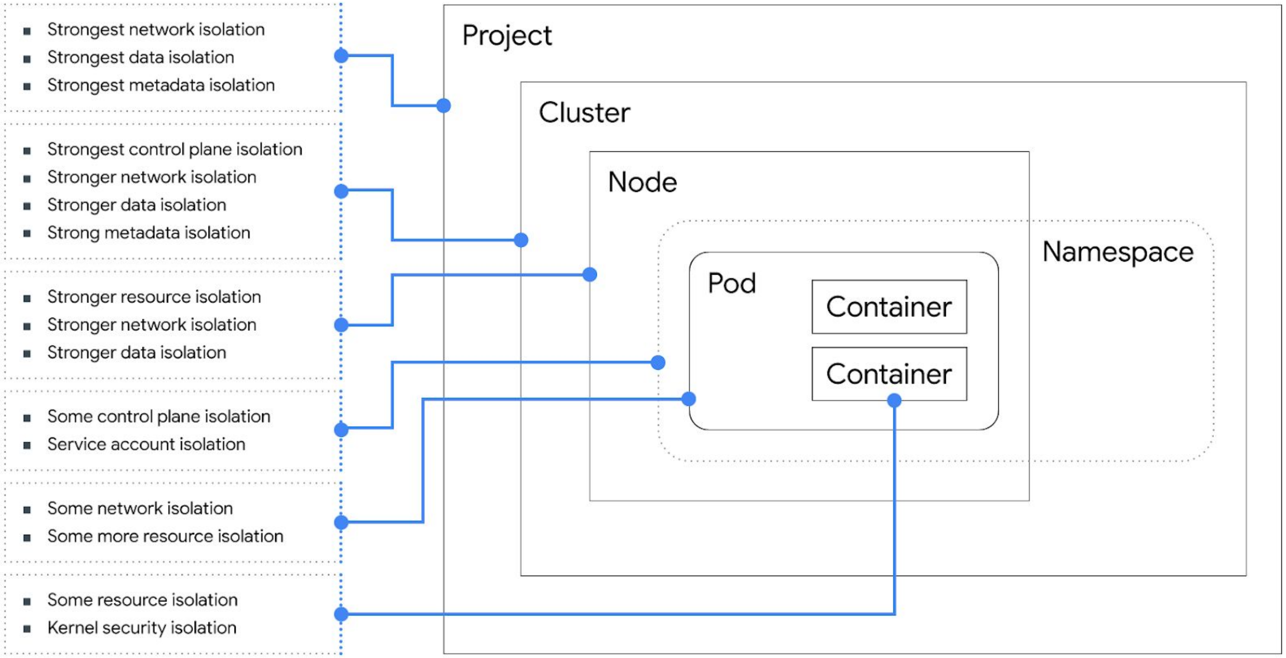
Kubernetes Security Overview



https://cloud.google.com/blog/products/gcp/exploring-container-security-isolation-at-different-layers-of-the-kubernetes-stack

Container

- Container Vulnerability Scanning and OS Dependency Security
- No root privileges
- Signature
- Isolation

Sandboxing / Container escape threat

Untrusted containers

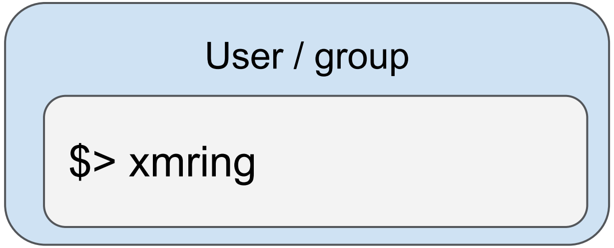
- **Root**
  - --privilege

```
$> xmring
```

Sandboxing / Container escape threat

Untrusted containers

- **No-Root**



Sandboxing / Container escape threat

Untrusted containers

- **No-Root**
- **Drop Capabilities**
    - CAP_SYS_ADMIN

Capabilities

User / group

$> xmring

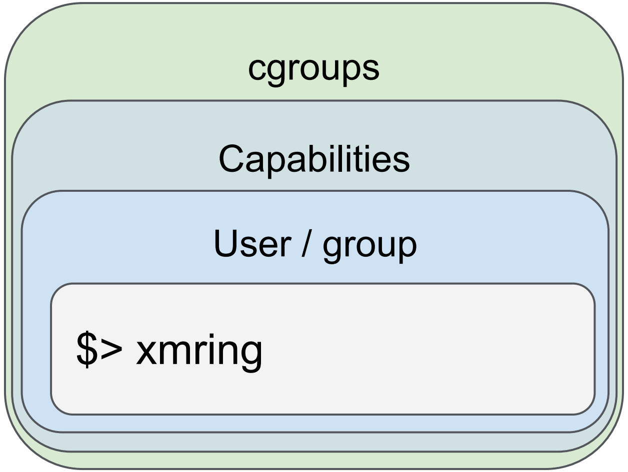
https://sysdig.com/blog/container-escape-capabilities-falco-detection/
https://man7.org/linux/man-pages/man7/capabilities.7.html

Sandboxing / Container escape threat

Untrusted containers

- No-Root
- Drop Capabilities

- Cgroups

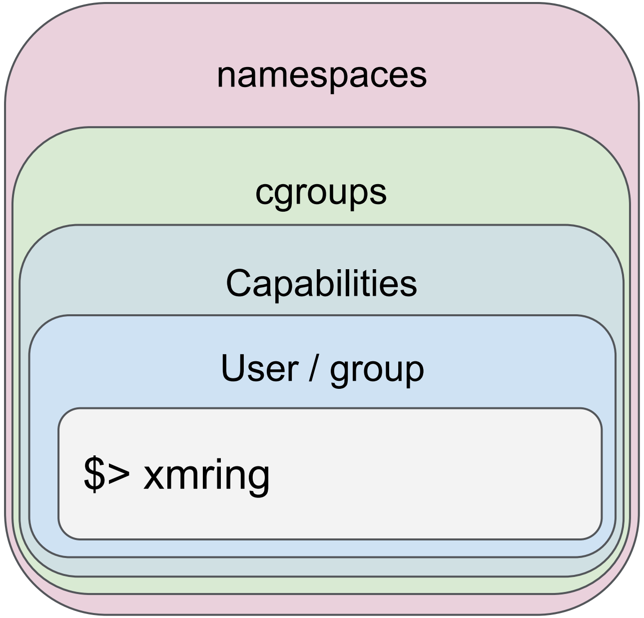


Sandboxing / Container escape threat

Untrusted containers

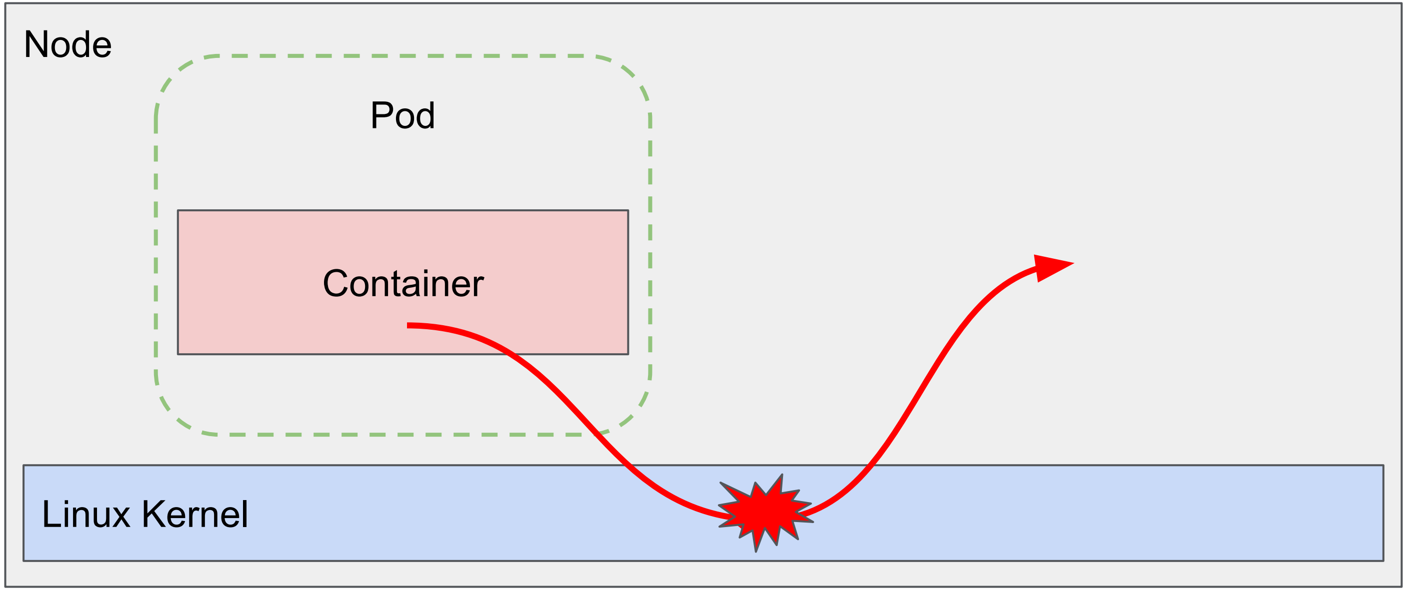
- No-Root
- Drop Capabilities

- Cgroups
- Namespaces



Sandboxing / Container escape threat

Untrusted containers



CVE-2022-0492, CVE-2022-0847 ,CVE-2019-5736

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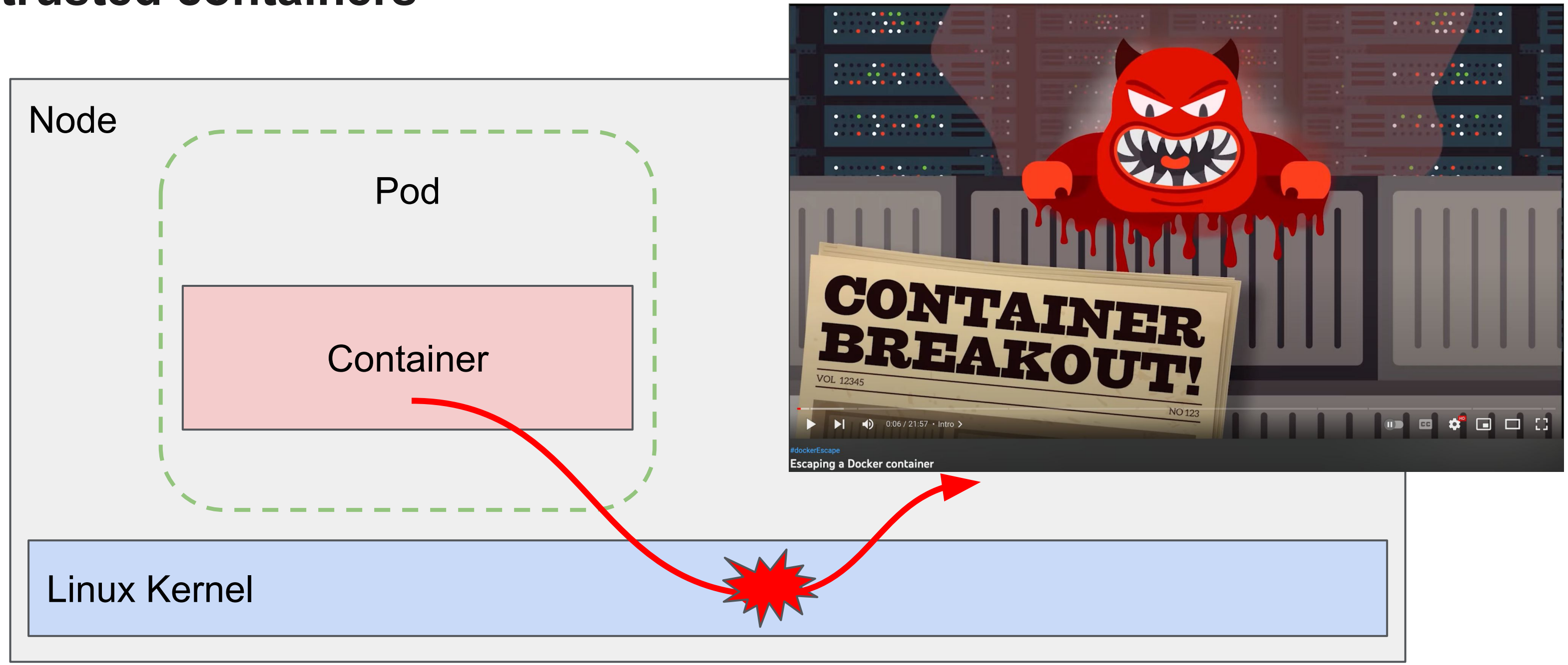
Sandboxing / Container escape threat

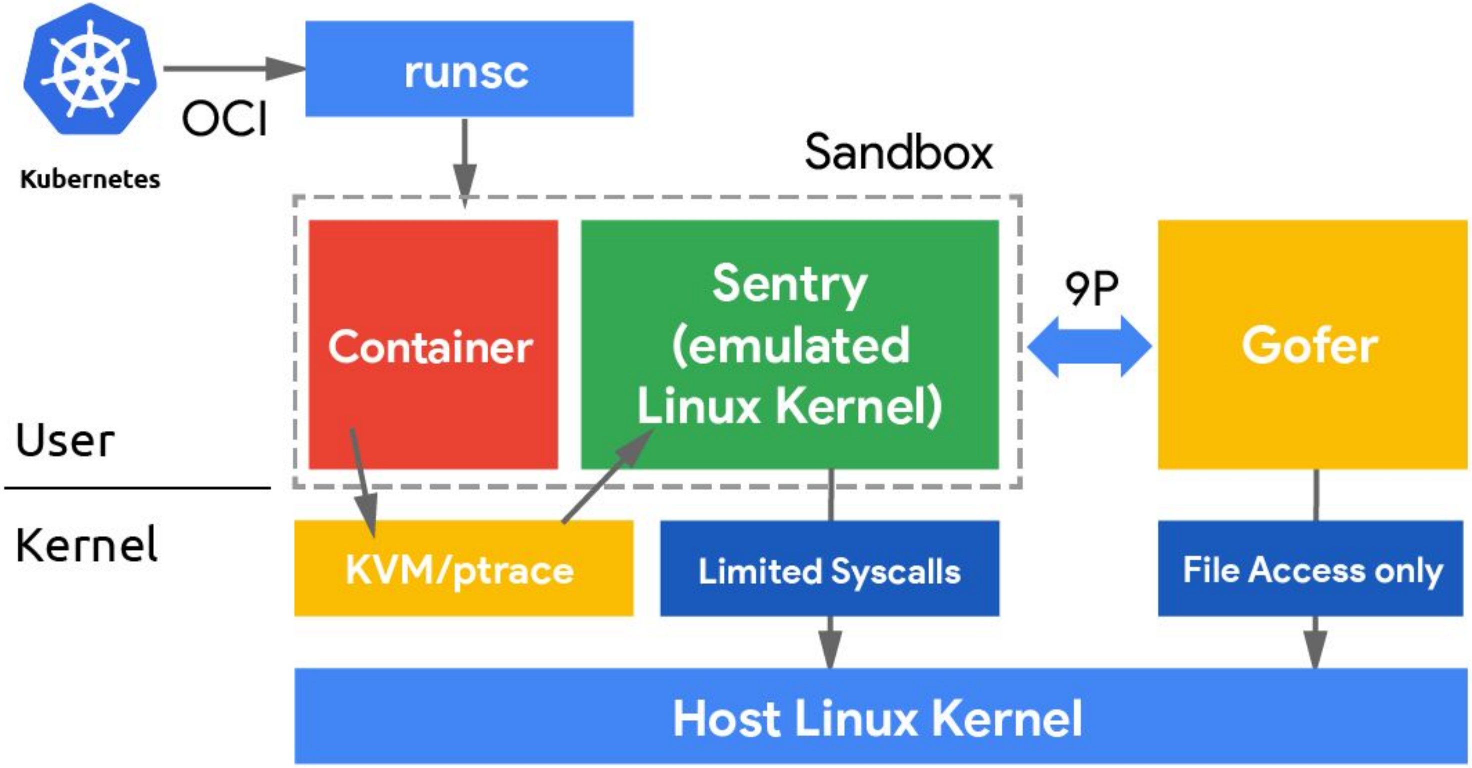
Untrusted containers

https://www.youtube.com/watch?v=ZbHVRoWEwA8



CVE-2022-0492, CVE-2022-0847 ,CVE-2019-5736

Gvisor



https://gvisor.dev/

Gvisor Docker

DEMO

https://gvisor.dev/docs/user_guide/install/

Gvisor Docker

```
ubuntu@ip-172-31-77-105:~$ docker run -it --runtime=runsc hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

ore ambitious, you can run an Ubuntu container with:
ubuntu bash

mate workflows, and more with a free Docker ID:
r.com/

and ideas, visit:
```

nstall/

```
ubuntu@ip-172-31-77-105:~$ docker run -it --runtime=runsc ubuntu dmesg
[    0.000000] Starting gVisor...
[    0.174391] Constructing home...
[    0.215607] Digging up root...
[    0.219148] Checking naughty and nice process list...
[    0.712065] Singleplexing /dev/ptmx...
[    0.749347] Accelerating teletypewriter to 9600 baud...
[    0.934577] Searching for socket adapter...
[    1.352682] Granting licence to kill(2)...
[    1.549927] Checking naughty and nice process list...
[    2.009177] Generating random numbers by fair dice roll...
[    2.036120] Searching for needles in stacks...
[    2.291776] Setting up VFS2...
[    2.457274] Ready!
```

Gvisor Minikube

DEMO

https://github.com/kubernetes/minikube/blob/master/deploy/addons/gvisor/README.md

Gvisor Minikube

```
ubuntu@ip-172-31-77-105:~$ kubectl get runtimeClass
NAME     HANDLER   AGE
gvisor   runsc     43h
ubuntu@ip-172-31-77-105:~$ kubectl apply -f untrusted-nginx.yaml
pod/nginx-untrusted created
ubuntu@ip-172-31-77-105:~$ cat untrusted-nginx.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-untrusted
spec:
  runtimeClassName: gvisor
  containers:
  - name: nginx
    image: nginx
ubuntu@ip-172-31-77-105:~$ kubectl get pods
NAME              READY   STATUS    RESTARTS   AGE
nginx-untrusted   1/1     Running   0          8s
```

```
ubuntu@ip-172-31-77-105:~$ minikube ssh
Last login: Wed Nov 16 12:45:40 2022 from 192.168.49.1
docker@minikube:~$ sudo -i
root@minikube:~# crictl pods
POD ID          CREATED          STATE      NAME                               NAMESPACE     ATTEMPT   RUNTIME
b3c134672ea42   10 seconds ago   Ready      nginx                              default       0         (default)
77a12fd235649   47 seconds ago   Ready      nginx-untrusted                    default       0         runsc
d6bea62b285ea   2 hours ago      Ready      storage-provisioner                kube-system   1         (default)
c81d19a6a9f02   2 hours ago      Ready      kindnet-ssztj                      kube-system   1         (default)
6ead1051be4c8   2 hours ago      Ready      coredns-565d847f94-5pqwn           kube-system   1         (default)
6dffa786646f3   2 hours ago      Ready      kube-apiserver-minikube            kube-system   1         (default)
e4bc73a5d1814   2 hours ago      Ready      etcd-minikube                      kube-system   1         (default)
95ee1b3d9d761   2 hours ago      Ready      kube-scheduler-minikube            kube-system   1         (default)
96b8a4158b848   2 hours ago      Ready      kube-controller-manager-minikube   kube-system   1         (default)
4b064c77027c0   44 hours ago     NotReady   coredns-565d847f94-5pqwn           kube-system   0         (default)
7bd5e0c668d7e   44 hours ago     NotReady   kube-proxy-z9jjh                   kube-system   0         (default)
e432078467a07   44 hours ago     NotReady   kindnet-ssztj                      kube-system   0         (default)
929797ee86dc2   44 hours ago     NotReady   kube-apiserver-minikube            kube-system   0         (default)
be0cbdf0b18e3   44 hours ago     NotReady   kube-controller-manager-minikube   kube-system   0         (default)
e391b9cb81a74   44 hours ago     NotReady   etcd-minikube                      kube-system   0         (default)
cf1aafc6b191b   44 hours ago     NotReady   kube-scheduler-minikube            kube-system   0         (default)
```

Gvisor -> Kubernetes (GKE)

```
root@cloud-client:~# kubectl get pods -n kube-system
NAME                                                       READY   STATUS    RESTARTS   AGE
event-exporter-gke-5dc976447f-kcvhh                        2/2     Running   0          54m
fluentbit-gke-52xc7                                        2/2     Running   0          53m
fluentbit-gke-jfjxb                                        2/2     Running   0          53m
fluentbit-gke-jjrb5                                        2/2     Running   0          53m
fluentbit-gke-jt2v6                                        2/2     Running   0          53m
fluentbit-gke-jvt85                                        2/2     Running   0
fluentbit-gke-kwhk6                                        2/2     Running   0
gke-metrics-agent-675wb                                    1/1     Running   0
gke-metrics-agent-6pqhs                                    1/1     Running   0
gke-metrics-agent-cjnsx                                    1/1     Running   0
gke-metrics-agent-kl5fm                                    1/1     Running   0
gke-metrics-agent-mkc7c                                    1/1     Running   0
gke-metrics-agent-qfhfp                                    1/1     Running   0
konnectivity-agent-669b59dcbc-2zgwg                        1/1     Running   0
konnectivity-agent-669b59dcbc-8xvw9                        1/1     Running   0
konnectivity-agent-669b59dcbc-f7qb4                        1/1     Running   0
konnectivity-agent-669b59dcbc-hpg56                        1/1     Running   0
konnectivity-agent-669b59dcbc-m6cvc                        1/1     Running   0
konnectivity-agent-669b59dcbc-xwscb                        1/1     Running   0
konnectivity-agent-autoscaler-658b588bb6-ssjdh             1/1     Running   0
kube-dns-autoscaler-fbc66b884-zwqbm                        1/1     Running   0
kube-dns-b8976dfcd-dp29c                                   4/4     Running   0
kube-dns-b8976dfcd-h6j6g                                   4/4     Running   0
kube-proxy-gke-cluster-demo-gvisor-default-pool-c0a0a60a-326q   1/1   Running   0
kube-proxy-gke-cluster-demo-gvisor-default-pool-c0a0a60a-7767   1/1   Running   0
kube-proxy-gke-cluster-demo-gvisor-default-pool-c0a0a60a-w4k5   1/1   Running   0
kube-proxy-gke-cluster-demo-gvisor-pool-gvisor-87f309f5-hsnt    1/1   Running   0
kube-proxy-gke-cluster-demo-gvisor-pool-gvisor-87f309f5-wllp    1/1   Running   0
kube-proxy-gke-cluster-demo-gvisor-pool-gvisor-87f309f5-xwz3    1/1   Running   0
l7-default-backend-6b99559c7d-zvtp7                        1/1     Running   0
metrics-server-v0.5.2-9b67f66b8-x7qgd                      2/2     Running   0          52m
pdcsi-node-9jpv2                                           2/2     Running   0          53m
pdcsi-node-9wb66                                           2/2     Running   0          53m
pdcsi-node-dxsr8                                           2/2     Running   0          53m
pdcsi-node-mpxnm                                           2/2     Running   0          53m
pdcsi-node-tqnlg                                           2/2     Running   0          53m
pdcsi-node-zs29h                                           2/2     Running   0          53m
```

```
root@cloud-client:~# cat httpd.yaml
# httpd.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: httpd
  labels:
    app: httpd
spec:
  replicas: 1
  selector:
    matchLabels:
      app: httpd
  template:
    metadata:
      labels:
        app: httpd
    spec:
      runtimeClassName: gvisor
      containers:
      - name: httpd
        image: httpd
root@cloud-client:~# kubectl create -f httpd.yaml --namespace default
deployment.apps/httpd created
root@cloud-client:~# kubectl get pods --namespace default -o wide
NAME                     READY   STATUS    RESTARTS   AGE   IP            NODE
busybox                  1/1     Running   0          69m   10.84.0.11    gke-cluster-demo-gvisor-default-pool-c0a0a60a
httpd-5df6d4b4f9-b8hv7   1/1     Running   0          11s   10.84.4.2     gke-cluster-demo-gvisor-pool-gvisor-87f309f5-
nginx                    1/1     Running   0          90m   10.84.0.10    gke-cluster-demo-gvisor-default-pool-c0a0a60a
root@cloud-client:~#
```

Gvisor -> Kubernetes (GKE)

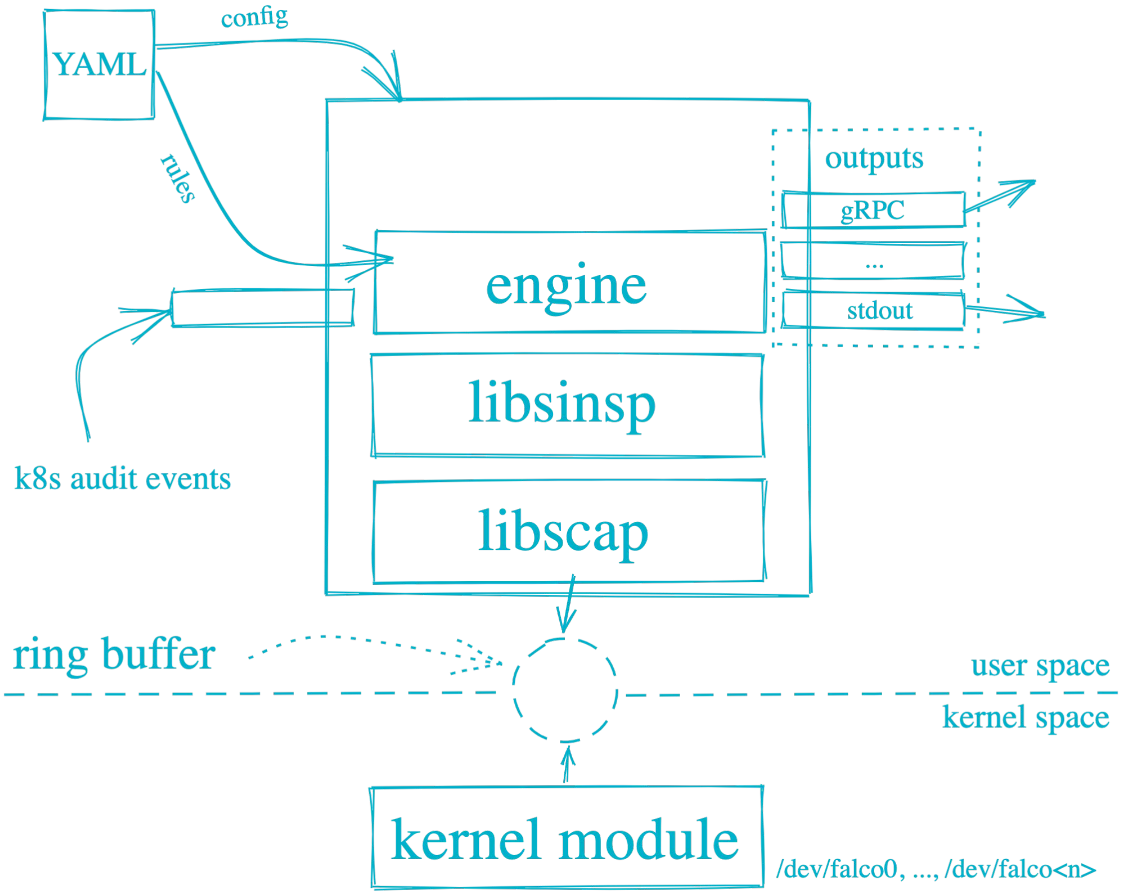
```
$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: compromised-pod
spec:
  containers:
  - name: compromised-container
    image: ghcr.io/datadog/dirtypipe-container-breakout
    imagePullPolicy: Always
    command: ["sh", "/wait-for-runc-and-overwrite.sh"]

$ kubectl apply -f pod.yaml
```

```
root@cloud-client:~# kubectl get pods --namespace default -o wide
NAME                      READY   STATUS    RESTARTS       AGE    IP           NODE
busybox                   1/1     Running   0              85m    10.84.0.11   gke-cluster-demo-gvisor-default-pool-c0a0a60a-w4k5
compromised-pod           0/1     Error     3 (33s ago)    65s    10.84.5.3    gke-cluster-demo-gvisor-pool-gvisor-87f309f5-xwz3
httpd-5df6d4b4f9-b8hv7    1/1     Running   0              15m    10.84.4.2    gke-cluster-demo-gvisor-pool-gvisor-87f309f5-hsnt
nginx                     1/1     Running   0              106m   10.84.0.10   gke-cluster-demo-gvisor-default-pool-c0a0a60a-w4k5
```

```
root@cloud-client:~# kubectl exec -it compromised-pod --namespace default -- /bin/bash
root@compromised-pod:/# ls
bin  boot  dev  etc  exploit  exploit.c  home  lib  lib32  lib64  libx32  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var  wait-for-runc-and-overwrite.sh
root@compromised-pod:/# ./wait-for-runc-and-overwrite.sh
Waiting for runC to be executed in the container
```

Falco



https://falco.org/

Falco Docker

DEMO

Falco Docker

```
ubuntu@ip-172-31-77-105:~$ docker run -ti ubuntu
root@b6ad0851bc4f:/# touch /bin/bash_fake
root@b6ad0851bc4f:/# cat /etc/shadow
root:*:19297:0:99999:7:::
daemon:*:19297:0:99999:7:::
bin:*:19297:0:99999:7:::
sys:*:19297:0:99999:7:::
sync:*:19297:0:99999:7:::
games:*:19297:0:99999:7:::
man:*:19297:0:99999:7:::
lp:*:19297:0:99999:7:::
mail:*:19297:0:99999:7:::
news:*:19297:0:99999:7:::
uucp:*:19297:0:99999:7:::
proxy:*:19297:0:99999:7:::
www-data:*:19297:0:99999:7:::
backup:*:19297:0:99999:7:::
list:*:19297:0:99999:7:::
irc:*:19297:0:99999:7:::
gnats:*:19297:0:99999:7:::
nobody:*:19297:0:99999:7:::
_apt:*:19297:0:99999:7:::
root@b6ad0851bc4f:/#
```

```
Wed Nov 16 14:05:21 2022: Loading rules from file /etc/falco/falco_rules.local.yaml
Wed Nov 16 14:05:21 2022: The chosen syscall buffer dimension is: 8388608 bytes (8 MB
s)
Wed Nov 16 14:05:21 2022: Starting health webserver with threadiness 2, listening on
port 8765
Wed Nov 16 14:05:21 2022: Enabled event sources: syscall
Wed Nov 16 14:05:21 2022: Opening capture with Kernel module
14:06:15.908766806: Notice A shell was spawned in a container with an attached termin
al (user=root user_loginuid=-1 pedantic_brahmagupta (id=b6ad0851bc4f) shell=bash pare
nt=<NA> cmdline=bash pid=349126 terminal=34816 container_id=b6ad0851bc4f image=ubuntu
)
14:06:26.857007567: Error File below a known binary directory opened for writing (use
r=root user_loginuid=-1 command=touch /bin/bash_fake pid=349159 file=/bin/bash_fake p
arent=bash pcmdline=bash gparent=<NA> container_id=b6ad0851bc4f image=ubuntu)
14:06:31.415912222: Warning Sensitive file opened for reading by non-trusted program
(user=root user_loginuid=-1 program=cat command=cat /etc/shadow pid=349162 file=/etc/
shadow parent=bash gparent=<NA> ggparent=<NA> gggparent=<NA> container_id=b6ad0851bc4
f image=ubuntu)
```

Falco Minikube

D

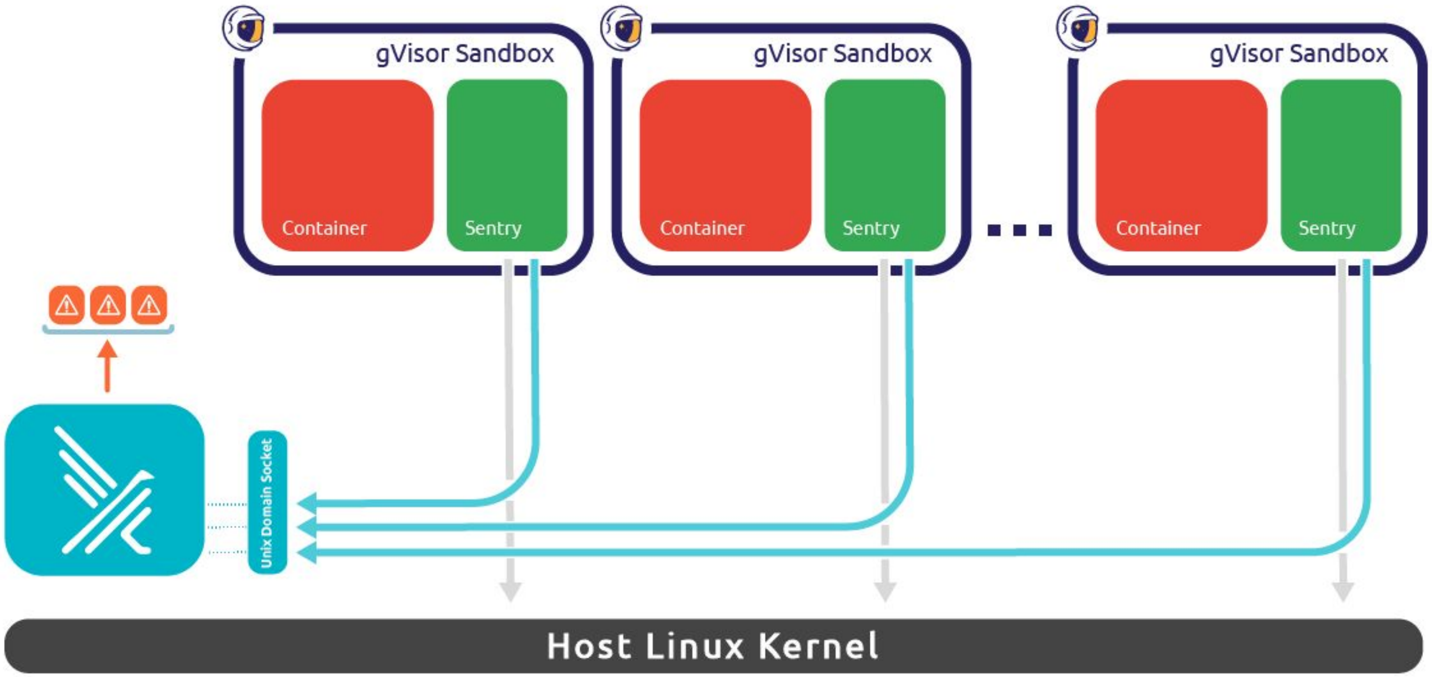
https:/

```
ubuntu@ip-172-31-72-232:~$ helm install falco falcosecurity/falco
NAME: falco
LAST DEPLOYED: Wed Nov 16 14:10:11 2022
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
Falco agents are spinning up on each node in your cluster. After a few
seconds, they are going to start monitoring your containers looking for
security issues.


No further action should be required.


Tip:
You can easily forward Falco events to Slack, Kafka, AWS Lambda and more with f
alcosidekick.
Full list of outputs: https://github.com/falcosecurity/charts/tree/master/falco
sidekick.
You can enable its deployment with `--set falcosidekick.enabled=true` or in you
r values.yaml.
See: https://github.com/falcosecurity/charts/blob/master/falcosidekick/values.y
aml for configuration values.
ubuntu@ip-172-31-72-232:~$ kubectl get pods
NAME          READY   STATUS     RESTARTS   AGE
falco-cjcvr   0/1     Init:0/1   0          5s
ubuntu@ip-172-31-72-232:~$ kubectl get pods
NAME          READY   STATUS     RESTARTS   AGE
falco-cjcvr   0/1     Init:0/1   0          8s
ubuntu@ip-172-31-72-232:~$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
falco-cjcvr   1/1     Running   0          8m11s
```

Gvisor and Falco -> Threat prevention and detection



Gvisor and Falco

DEMO

https://falco.org/blog/intro-gvisor-falco/
https://gvisor.dev/docs/tutorials/falco/

Gvisor and Fa

```
14:24:25.375030341: Notice Unexpected setuid call by non-sudo, non-roo
t program (user=<NA> user_loginuid=-1 cur_uid=65534 parent=runsc comma
nd=exe --root=/var/run/docker/runtime-runc/moby --log=/run/containerd/
io.containerd.runtime.v2.task/moby/fc5ed14a11623664029d170b7e0a4ff9613
4ec741637bc30d9e1573e30519164/log.json --log-format=json --systemd-cgr
oup=true --pod-init-config=/etc/docker/runsc_falco_config.json --log-f
d=3 boot --bundle=/run/containerd/io.containerd.runtime.v2.task/moby/f
```

```
ubuntu@ip-172-31-77-105:~$ docker run -ti --runtime=runsc ubuntu
root@fc5ed14a1162:/# touch /bin/bash_fake
root@fc5ed14a1162:/# exit
exit
ubuntu@ip-172-31-77-105:~$
```

```
ge=<NA>)
14:24:33.321975940: Error File below a known binary directory opened f
or writing (user=nobody user_loginuid=-1 command=exe pid=349676 file=/
bin/bash_fake parent=exe pcmdline=exe gparent=exe container_id=fc5ed14
a1162 image=ubuntu)
14:24:34.830325071: Warning a shell configuration file has been modifi
ed (user=nobody user_loginuid=-1 command=exe pid=349664 pcmdline=exe f
ile=/root/.bash_history container_id=fc5ed14a1162 image=ubuntu)
14:24:34.830482147: Warning a shell configuration file has been modifi
ed (user=nobody user_loginuid=-1 command=exe pid=349664 pcmdline=exe f
ile=/root/.bash_history container_id=fc5ed14a1162 image=ubuntu)
```

Gvisor and Falco

```
ubuntu@ip-172-31-77-105:~$ sudo falco   -c /etc/falco/falco.yaml   --g
visor-config /etc/falco/pod-init.json   --gvisor-root /var/run/docker/
runtime-runc/moby
Wed Nov 16 14:26:59 2022: Falco version: 0.33.0 (x86_64)
Wed Nov 16 14:26:59 2022: Falco initialized with configuration file: /
etc/falco/falco.yaml
```

```
ubuntu@ip-172-31-77-105:~$ docker run -ti --runtime=runsc ubuntu
root@6bee6b65c801:/# touch /bin/bash_fake_gvisor
root@6bee6b65c801:/# cat /etc/shadow
root:*:19297:0:99999:7:::
daemon:*:19297:0:99999:7:::
bin:*:19297:0:99999:7:::
```

```
Wed Nov 16 14:26:59 2022: Enabled event sources: syscall
Wed Nov 16 14:26:59 2022: Opening capture with gVisor. Configuration p
ath: /etc/falco/pod-init.json
14:27:07.302126396: Error File below a known binary directory opened f
or writing (user=root user_loginuid=0 command=touch touch /bin/bash_fa
ke_gvisor pid=<NA> file=/bin/bash_fake_gvisor parent=bash pcmdline=bas
h bash gparent=<NA> container_id=6bee6b65c801 image=ubuntu) container=
exciting_raman (id=6bee6b65c801) vpid=4 vtid=4
14:27:14.528463977: Warning Sensitive file opened for reading by non-t
rusted program (user=root user_loginuid=0 program=cat command=cat cat
/etc/shadow pid=<NA> file=/etc/shadow parent=bash gparent=<NA> ggparen
t=<NA> gggparent=<NA> container_id=6bee6b65c801 image=ubuntu) containe
r=exciting_raman (id=6bee6b65c801) vpid=5 vtid=5
14:27:16.246792932: Warning Shell history had been deleted or renamed
(user=<NA> user_loginuid=0 type=open command=bash bash pid=<NA> fd.nam
e=/root/.bash_history name=/root/.bash_history path=<NA> oldpath=<NA>
container=exciting_raman (id=6bee6b65c801) vpid=1 vtid=1)
```

D

https://fa

https://

Gvisor and Falco -> Kubernetes (minikube)

DEMO

Gvisor and Falco -> K

```
ubuntu@ip-172-31-77-105:~$ sudo falco   -c /etc/falco/falco.yaml   --g
visor-config /etc/falco/minikube-pod-init.json
Wed Nov 16 14:29:55 2022: Falco version: 0.33.0 (x86_64)
Wed Nov 16 14:29:55 2022: Falco initialized with configuration file: /
etc/falco/falco.yaml
Wed Nov 16 14:29:55 2022: Loading rules from file /etc/falco/falco_rul
```

```
ubuntu@ip-172-31-77-105:~$ kubectl apply -f untrusted-nginx.yaml
pod/nginx-untrusted created
ubuntu@ip-172-31-77-105:~$ kubectl exec -it nginx-untrusted -- bash
root@nginx-untrusted:/# touch /bin/bash-fake-gvisor-minikube
root@nginx-untrusted:/# exit
exit
```

```
tc/falco/falco_rul

dimension is: 8388

with threadiness 2

call
r. Configuration p
```

D

```
14:30:41.103057346: Error File below a known binary directory opened f
or writing (user=<NA> user_loginuid=0 command=touch touch /bin/bash-fa
ke-gvisor-minikube pid=<NA> file=/bin/bash-fake-gvisor-minikube parent
= pcmdline= gparent=<NA> container_id=2507520f23e2 image=<NA>) contain
er=<NA> (id=2507520f23e2) vpid=28 vtid=28
14:30:44.244179083: Warning a shell configuration file has been modifi
ed (user= user_loginuid=-1 command= pid=<NA> pcmdline=<NA> file=/root/
.bash_history container_id=host image=<NA>) container=host (id=host) v
pid=<NA> vtid=<NA>
14:30:44.244343062: Warning a shell configuration file has been modifi
ed (user= user_loginuid=-1 command= pid=<NA> pcmdline=<NA> file=/root/
.bash_history container_id=host image=<NA>) container=host (id=host) v
pid=<NA> vtid=<NA>
```

```
ubuntu@ip-172-31-77-105:~$ cat untrusted-nginx.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-untrusted
spec:
  runtimeClassName: gvisor
  containers:
  - name: nginx
    image: nginx
```

Conclusions

- **Secure your Cloud provider with Principle of least privilege.**
- **Secure your Cluster Kubernetes**
- **Secure the Host / OS**
- **Secure the containers**
    - **Isolation**
        - **Sandboxing**
            - **Gvisor**
- **Secure your code**
- **Threat detection with Falco**

Reforzando la seguridad de Kubernetes con gVisor y Falco

16 Nov

17:00

Miguel Hernández

Sysdig

KCD SPAIN